

Modelling of Stochastic Processes

Gytis Vejelis

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1 Equations Of Motion

From Apostolos derivations we have:

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}}) = 0 \quad (1)$$

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]\tilde{\chi}_{\mathbf{k}} = 0 \quad (2)$$

$$\langle\chi^2\rangle\ddot{\tilde{\theta}_{\mathbf{k}}} + (2\langle\chi\dot{\chi}\rangle + 3H\langle\chi^2\rangle)\dot{\tilde{\theta}_{\mathbf{k}}} + \frac{k^2}{a^2}\langle\chi^2\rangle\tilde{\theta}_{\mathbf{k}} = 0 \quad (3)$$

where the averages are

$$\langle\chi^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\tilde{\chi}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

$$\langle\dot{\theta}^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\dot{\tilde{\theta}}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

I want to rewrite these equations in a "vector" form for numerical integration, such that for instance

$$x = \tilde{\chi}_{\mathbf{k}} \quad \text{and} \quad v = \dot{\tilde{\chi}_{\mathbf{k}}}$$

so that

$$X = \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} \tilde{\chi}_{\mathbf{k}} \\ \dot{\tilde{\chi}_{\mathbf{k}}} \end{bmatrix}$$

Then

$$\dot{v} + 3Hv + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]x = 0$$

To get a better understanding of these equations I will first compact some terms. Changing (1) into:

$$\ddot{\phi} + 3H\dot{\phi} - V(\phi, \langle\chi\rangle) = 0$$

with

$$V(\phi, \langle\chi^2\rangle) = \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}})$$

Then (2)

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \omega\tilde{\chi}_{\mathbf{k}} = 0$$

$$\omega = \omega(\phi, \langle\chi^2\rangle, \langle\dot{\theta}\rangle, \langle(\nabla\theta)^2\rangle) = \frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle$$

and for (3) I am pretty sure I can just divide by $\langle\chi^2\rangle$

$$\ddot{\tilde{\theta}_{\mathbf{k}}} + \left(2\frac{\langle\chi\dot{\chi}\rangle}{\langle\chi^2\rangle} + 3H\right)\dot{\tilde{\theta}_{\mathbf{k}}} + \frac{k^2}{a^2}\tilde{\theta}_{\mathbf{k}} = 0$$

To make this managable computable, I think I need to first compute the averages of all.

2 Simpler equations

It makes sense to first start simulations with simpler equations where we take $\langle \dot{\theta} \rangle = \langle (\nabla \theta)^2 \rangle = 0$ and forget about the third equation entirely.

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp \left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] + g^2 \langle \chi^2 \rangle (\phi - \phi_{\text{SBP}}) = 0 \quad (4)$$

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda \langle \chi^2 \rangle \right] \tilde{\chi}_{\mathbf{k}} = 0 \quad (5)$$

where we can go a step further and first consider the homogeneous equation

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp \left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] = 0 \quad (6)$$

To compute the averages we must discretise the integral over $\mathbf{k}$ modes, and first we will not consider any 2PI corrections

$$\langle \chi^2 \rangle = \int_{\mathbb{R}^3} \frac{d^3 p}{(2\pi)^3} |\tilde{\chi}_{\mathbf{p}}(t)|^2 \approx \sum_{\mathbf{p}} \omega_{\mathbf{p}} |\tilde{\chi}_{\mathbf{p}}(t)|^2$$

where $\omega_{\mathbf{p}}$ being the weights. There are several choices of $\omega_{\mathbf{p}}$, but since $\langle \chi^2 \rangle$ is not isometric we use

$$\omega_{\mathbf{k}} = \frac{\delta k_x \delta k_y \delta k_z}{(2\pi)^3}$$

which is determined at the start of the simulation when we choose the change of k , assuming that the range is divided in equal steps that is $\delta k_x = \delta k_y = \delta k_z = \delta k \equiv \text{const.}$

$$\omega_{\mathbf{k}} = \left(\frac{\delta k}{2\pi} \right)^3 = \omega$$

yielding the simplified expression

$$\langle \chi^2 \rangle \approx \omega \sum_{\mathbf{p}} |\tilde{\chi}_{\mathbf{p}}(t)|^2$$

Another important thing to note is that the hubble parameter H also changes, since it depends on the energy density ρ of the fields

$$\begin{aligned} \rho_{\phi}(t) &= \frac{1}{2} \dot{\phi}^2 + V(\phi) \\ \rho_{\chi}(t) &= \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2} (|\dot{\tilde{\chi}_{\mathbf{k}}}|^2 + \omega_k) \end{aligned}$$

Then $\rho_{\text{tot}} = \rho_{\phi} + \rho_{\chi}$, however at first we can neglect the ρ_{χ} term and the hubble parameter thus becomes

$$H \approx \sqrt{\frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right)} = \sqrt{\frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] \right)}$$

Note that we are using the potential $V(\phi) = V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right]$.

Next, I need to determine initial conditions. It makes sense to integrate from the symmetry breaking point ϕ_{SBP} . During kination (point where energy density is dominated by kinetic energy) the ϕ field equation simplifies immensely:

$$\ddot{\phi} + 3H\dot{\phi} = 0$$

with a simplified Hubble parameter $H \approx \frac{|\dot{\phi}|}{\sqrt{6}M_{\text{pl}}}$. We can then solve these equations. Note that we will take $t_i = t_{\text{SBP}} = 0$ and $v = \dot{\phi}_{\text{SBP}}$

$$\begin{aligned}\phi &= \sqrt{\frac{2}{3}} \text{sign}(v) \ln \left(1 + \sqrt{\frac{3}{2}} \frac{|v|t}{M_{\text{pl}}} \right) M_{\text{pl}} \\ \dot{\phi} &= v \exp \left(-\sqrt{\frac{3}{2}} \frac{\phi}{M_{\text{pl}} \text{sign}(v)} \right)\end{aligned}$$

That means that initial conditions $\phi_0 = \phi(t_i) = 0$ and $\dot{\phi}_0 = \dot{\phi}(0) = v$, where I think I can choose v - being a free parameter. We can similarly consider modes for χ field. For large $|\phi|$ they satisfy

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \omega_k^2 \chi_k = 0$$

where $\omega^2 = k^2 + g^2 \phi^2 - \lambda f^2$. The initial conditions are given by the Bunch-Davies vacuum state

$$\chi_{\phi, \text{vac}} = \frac{a^{-1}}{\sqrt{2\omega_k}} \exp \left\{ -i \int^t \omega_k dt' \right\}$$

But again, since our initial conditions have $t = 0$ the above simplifies to just:

$$\chi_k(t_i) = \frac{1}{a(t_i)\sqrt{\omega_k}} = \frac{1}{\sqrt{\omega_k}}$$

Where we have used $a(t_i) = 1$. We can also differentiate χ_k modes to obtain initial conditions for $\dot{\chi}_k$, where we will only keep the adiabatic terms:

$$\dot{\chi} = [-H(t_i) - i\omega_k(t_i)]\chi_k(t_i) \approx 0$$

All of these initial conditions are dependent on parameters g, λ, f, v and technically t_i , though we set it to 0 previously. We can discuss the ranges of these parameters:

1. $10^{-5} < \lambda < 1$
2. $f = 5 \times 10^{-7}$
3. $10^{-14} < g < 1$
4. v is free

3 Integration strategy

First things first I need to change the variables from time to number of e-folds

$$t \rightarrow N, \quad \text{where} \quad N = \ln a$$

Then

$$\frac{dN}{dt} = \frac{\dot{a}}{a} = H \quad \Rightarrow \quad \frac{d\phi}{dt} = H \frac{d\phi}{dN} \quad \text{and} \quad \frac{d^2\phi}{dt^2} = H \frac{dH}{dN} \frac{d\phi}{dN} + H^2 \frac{d^2\phi}{dN^2}$$

In prime notation for any function A

$$\dot{A} = H A' \quad \text{and} \quad \ddot{A} = H^2 A'' + H' H A'$$

Then using Klein-Gordon, Friedman and acceleration equations

$$\begin{aligned} \ddot{\phi} + 3H\dot{\phi} + V_{,\phi} &= 0 \\ H^2 &= \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2}\dot{\phi}^2 + V(\phi) \right) \\ \dot{H} &= -\frac{1}{2M_{\text{pl}}^2} \dot{\phi}^2 \end{aligned}$$

We rewrite the derivatives

$$\begin{aligned} H^2\phi'' + H'H\phi' + 3H^2\phi' + V_{,\phi} &= 0 \\ H^2 &= \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2}H^2\phi'^2 + V(\phi) \right) \\ HH' &= -\frac{1}{2M_{\text{pl}}^2} H^2\phi' \end{aligned}$$

The last two equations can be rewritten in a more convenient form: in the first I'm just expressing H^2 in the second I'm expressing H'/H , which are the terms that will appear when we rearrange KG

$$H^2 = \frac{V(\phi)}{3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2} \quad \text{and} \quad \frac{H'}{H} = -\frac{1}{2M_{\text{pl}}^2} \phi'^2$$

Then after dividing KG by H^2 and inserting these relations I obtain

$$\phi'' + \left(3 - \frac{\phi'^2}{2M_{\text{pl}}^2} \right) \phi' + \left(3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2 \right) \frac{V_{,\phi}}{V} = 0$$

I'm not sure if I can do this, but I noticed that

$$W = \ln V \quad \Rightarrow \quad W_{,\phi} = \frac{V_{,\phi}}{V}$$

This is especially useful, since our potential is exponential, so using $V(\phi) = V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right]$ we get

$$W = -\kappa \exp\left(\frac{\phi}{bM_{\text{pl}}}\right) + W_0 \quad \text{and} \quad W_{,\phi} = -\frac{\kappa}{bM_{\text{pl}}} \exp\left(\frac{\phi}{bM_{\text{pl}}}\right)$$

where $W_0 = \ln V_0$.

For convenience I'll rewrite everything more explicitly with $M_{\text{pl}} = 1$

$$\phi'' + \left(3 - \frac{\phi'^2}{2}\right) \phi' + \left(3 - \frac{1}{2}\phi'^2\right) \left(-\frac{\kappa}{b} \exp\left(\frac{\phi}{b}\right)\right) = 0$$

There is further massaging I can do, but for now I can also integrate this equation as is. Take $u_1 = \phi$ and $u_2 = \phi'$, then

$$\begin{aligned} u'_1 &= u_2 \\ u'_2 &= \left(\frac{1}{2}u_2^2 - 3\right) u_2 + \left(3 - \frac{1}{2}u_2^2\right) \frac{\kappa}{b} \exp\left(\frac{u_1}{b}\right) \end{aligned}$$

I can further masage this equation (Not sure if its worth to), by multiplying everything by ϕ' , then $\phi''\phi' = \frac{1}{2}(\phi'^2)'$ and $W_{,\phi}\phi' = bM_{\text{pl}}W'$, hence

$$\frac{1}{2}(\phi'^2)' + \left(3 - \frac{\phi'^2}{2M_{\text{pl}}^2}\right) \phi'^2 + \left(3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2\right) bM_{\text{pl}}W' = 0$$

Taking $M_{\text{pl}} = 1$ and multiplying everything by 2

$$(\phi'^2)' + (6 - \phi'^2) \phi'^2 + (6 - \phi'^2) bW' = 0$$

Another approach, instead of multiplying everything by ϕ' we can just split these equations into a system with $u = \phi'$, then

$$\begin{cases} u = \phi' \\ u' + (6 - u^2)(u^2 + bW_{,\phi}) = 0 \end{cases}$$

Since I will be solving these equations numerically, I need some analytic results to check my calculations with. I will use slow-roll and kination approximations.

For slow-roll we consider that $|\ddot{\phi}| \ll |3H\dot{\phi}|$ and $|\dot{\phi}| \ll |V_{,\phi}(\phi)|$, so Friedman equation simplifies to

$$H^2 = \frac{V(\phi)}{3M_{\text{pl}}^2}$$

And Slow-roll klein gordon (wrt time) equation becomes:

$$3H\dot{\phi} = -V_{,\phi}(\phi)$$

Again, using the same change of variables $t \rightarrow N$ we have

$$3H^2\phi' = -V_{,\phi} \quad \Rightarrow \quad \phi' = -M_{\text{pl}}^2 \frac{V_{,\phi}(\phi)}{V(\phi)} = -M_{\text{pl}}^2 W_{,\phi}$$

This now should be solvable analytically, again, setting $M_{\text{pl}} = 1$

$$N = \int -\frac{1}{W_{,\phi}} d\phi = \int \frac{b}{\kappa} \exp\left(-\frac{\phi}{b}\right) d\phi = -\frac{b^2}{\kappa} \exp\left(-\frac{\phi}{b}\right)$$

There is also the kination limit, which was discussed a bit earlier.

4 End of inflation

Note that inflation lasts until $\epsilon = 1$, where

$$\epsilon = -\frac{\dot{H}}{H^2} \quad \text{or} \quad \epsilon = \frac{\dot{\phi}^2}{2M_{\text{pl}}^2 H^2}$$

where for inflaton field we use $\rho_\phi = T^{00} = \frac{1}{2}\dot{\phi}^2 + V(\phi)$, assuming that the field is homogeneous, then $\nabla\phi = 0$. But this is not that necessary since we can change the integration coordinates into number of e-folds, also taking $\epsilon = 1$ and $M_{\text{pl}} = 1$, we get

$$\dot{\phi}^2 = 2H^2 \quad \Rightarrow \quad H^2(\phi')^2 = 2H^2 \quad \Rightarrow \quad \phi' = \pm\sqrt{2}$$

which means that at the end of inflation $\phi' = \sqrt{2}$ and in principle I can integrate backwards.

January

Notes from end of January

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I want to apply slowroll approximation to the inflaton evolution and only then change variables to the number of efolds. I had this solved numerically previously, but I think I accidentally removed it. Either way it's good to do it again and compare numerical calculations with analytic calculations.

In slow-roll regime kinetic energy is negligible compared to the potential and acceleration is negligible

$$\dot{\phi} \ll V(\phi) \quad \text{and} \quad |\ddot{\phi}| \ll |3H\dot{\phi}|$$

Then Friedmann equation simplifies to

$$H^2 = \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right) \rightarrow H^2 = \frac{V(\phi)}{3M_{\text{pl}}^2}$$

and the equation of motion for the inflaton simplifies to

$$3H\dot{\phi} + V_{,\phi}(\phi) = 0$$

Now switch integration variables.

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Switching the integration variable to the number of efolds

$$3H^2\phi' = -V_{,\phi} \Rightarrow \phi' = -\frac{V_{,\phi}(\phi)}{V(\phi)}$$

where we used $M_{\text{pl}} = 1$. With $V(\phi) = V_0 \exp \left[-\kappa e^{\frac{\phi}{b}} \right]$ and $V(\phi)_{,\phi} = V(\phi) \frac{-\kappa}{b} \exp \left(\frac{\phi}{b} \right)$ we're left with

$$\phi' = \frac{\kappa}{b} \exp \left(\frac{\phi}{b} \right) \Rightarrow \int_{\phi_i}^{\phi_f} \frac{b}{\kappa} \exp \left(-\frac{\phi}{b} \right) d\phi = N_i - N_f = \Delta N$$

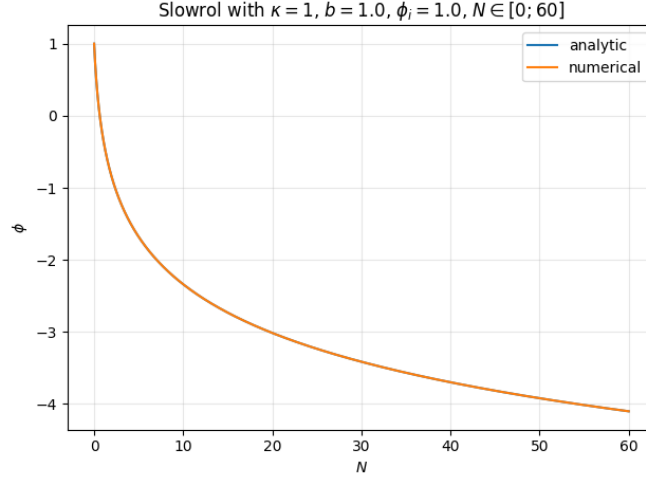
Integrating and inverting to obtain phi, we have

$$\Delta N = \frac{b^2}{\kappa} \left(e^{-\phi_i/b} - e^{-\phi_f/b} \right) \Rightarrow \phi_f = -b \ln \left(e^{-\phi_i/b} - \frac{\kappa}{b^2} \Delta N \right)$$

At the start of inflation, by definition $N_i = 0$ (if we define $N(t) \equiv \ln \frac{a(t)}{a(t_i)}$), we have then that slow-roll up until inflation end looks like:

$$\phi(N) = -b \ln \left(e^{-\phi_i/b} + \frac{\kappa}{b^2} N \right) \tag{7}$$

Implemented the numerical solution and compared with analytic solution:



They are so extremely close you can't even see the analytic solution. Next, I want to learn a bit about perturbations. I've found slides "Cosmic Inflation - Mini Lecture" by Jinsu Kim

They say that once the scalar potential $V(\phi)$ is given, then you can compute cosmological observables

- Calculate slow-roll parameters:

$$\epsilon = \frac{M_{\text{pl}}^2}{2} \left(\frac{V_{,\phi}}{V} \right)^2, \quad \text{and} \quad \eta = M_{\text{pl}}^2 \frac{V_{,\phi\phi}}{V}$$

- Find the end of inflation $\epsilon \approx 1$
- Find the horizon-crossing point $N = 60$ (i.e. CMB scale)

$$N = \int H dt, \quad H^2 \approx \frac{V}{3M_{\text{pl}}^2}$$

- Compute cosmological observables at the horizon-crossing point:

$$\mathcal{P}_s = \frac{V}{24\pi^2 M_{\text{pl}}^4 \epsilon}, \quad n_s = 1 + 6\epsilon + 2\eta, \quad r = 16\epsilon$$

We know that there were perturbations at some point in the early universe due to the large scale structures we observe today, hence we need to spoil homogeneity and isotropy of the Universe. Assume that the inflaton field ϕ is decomposed as

$$\phi(t, \mathbf{x}) = \varphi(t) + \delta\phi(t, \mathbf{x})$$

where $\varphi(t)$ is the background field, only time dependent, $\delta\phi(t, \mathbf{x})$ is the perturbation (function of spacetime). The action is given by

$$S_{EH} = \frac{M_{\text{pl}}^2}{2} \int d^4x \sqrt{-g} R, \\ S_\phi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \quad (8)$$

with $S = S_{EH} + S_\phi$. The equations of motion for the inflaton is obtained by varying the action wrt ϕ (we treat $g_{\mu\nu}$ as fixed).

$$\delta S_\phi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \delta(\partial_\mu \phi \partial_\nu \phi) - V_{,\phi}(\phi) \delta\phi \right]$$

Then using the product rule and $\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi)$

$$\delta(\partial_\mu \phi \partial_\nu \phi) = (\partial_\mu \delta\phi) \partial_\nu \phi + \partial_\mu \phi (\partial_\nu \delta\phi)$$

$$-\frac{1}{2} g^{\mu\nu} (\partial_\mu \delta\phi) \partial_\nu \phi - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi (\partial_\nu \delta\phi)$$

but if we switch $\mu \leftrightarrow \nu$ on the second term and commute then we the same thing. Yielding

$$\delta S = \int d^4x \sqrt{-g} [-g^{\mu\nu} \partial_\mu \phi \partial_\nu (\delta\phi) - V']$$

As is usual in these sort of variations we integrate by parts and the boundary term vanished. In order to have the correct integration measure we also factor out $\sqrt{-g}$ hence

$$\delta S = \int d^4x \delta\phi [\partial_\nu (\sqrt{-g} g^{\mu\nu} \delta_\mu \phi) - \sqrt{-g} V_{,\phi}(\phi)] = \int d^4x \sqrt{-g} \delta\phi \left[\frac{1}{\sqrt{-g}} \partial_\nu (\sqrt{-g} g^{\mu\nu} \delta_\mu \phi) - \sqrt{-g} V_{,\phi}(\phi) \right]$$

and finally $\delta S = 0$ for arbitrary $\delta\phi$, hence we obtain the Euler-Lagrange equation and in principle we are done:

$$\frac{1}{\sqrt{-g}} \partial_\nu (\sqrt{-g} g^{\mu\nu} \delta_\mu \phi) - \sqrt{-g} V_{,\phi}(\phi) = 0.$$

However, we can remember that the divergence of a vector $\nabla_\mu V^\mu = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} V^\mu)$, but $\nabla^\mu \phi = \partial^\mu \phi = g^{\mu\nu} \partial_\nu \phi$ defines a vector, hence

$$\square\phi \equiv \nabla_\mu \nabla^\mu \phi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi) \quad (9)$$

where the $\square$ is the curved d'Alembertian. Leaving the Klein-Gordon equation

$$\square\phi - V'(\phi) = 0.$$

Again, in principle we are done, however for cosmology we use FLRW metric with homogeneous $\phi(t)$ and note that FLRW assumes isotropy hence we use spherical coordinates

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\varphi^2) \right],$$

$$g_{\mu\nu} = \text{diag} \left(-1, \frac{a^2}{1 - kr^2}, a^2 r^2, a^2 r^2 \sin^2 \theta \right),$$

$$g = \det(g_{\mu\nu}) = (-1) \cdot \frac{a^2}{1 - kr^2} \cdot a^2 r^2 \cdot a^2 r^2 \sin^2 \theta = -a^6 r^4 \sin^2 \theta.$$

This now allows us to compute $\square$, using equation (9)

$$\begin{aligned}
\Box\phi &= \frac{1}{\sqrt{a^6 r^4 \sin^2 \theta}} \partial_\mu (\sqrt{a^6 r^4 \sin^2 \theta} g^{\mu t} \partial_t \phi) \\
&= \frac{1}{a^3} \partial t (a^3 \dot{\phi}) \\
&= \frac{1}{a^3} [3a^2 \dot{a} \dot{\phi} + a^3 \ddot{\phi}] \\
&= \ddot{\phi} + 3H \dot{\phi}
\end{aligned}$$

where $H = \frac{\dot{a}}{a}$ is the hubble parameter. Finally giving the Klein-Gordon equation that we know

$$\ddot{\phi} + 3H \dot{\phi} + V'(\phi) = 0$$

Next we take a general expression for the energy-momentum tensor

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_\phi}{\delta g^{\mu\nu}}$$

Note that to compute a general functional derivative $\frac{\delta S}{\delta X}$ we use

$$\delta S = \int d^4x \frac{\delta S}{\delta X(x)} \delta X(x).$$

Notice that there is no integration measure $\sqrt{-g}!$ Roughly speaking, the energy momentum tensor is just a special case with the functional derivative being wrt the metric. So we again vary the action (8), but this time wrt the metric.

$$\begin{aligned}
\delta S_\phi &= \int d^4x \sqrt{-g} \delta(\sqrt{-g}) \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] + \sqrt{-g} \left[-\frac{1}{2} \delta(g^{\mu\nu}) \partial_\mu \phi \partial_\nu \phi \right] \\
&= \int d^4x \left\{ \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \right) \left[-\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right] + \sqrt{-g} \left[-\frac{1}{2} \partial_\mu \phi \partial_\nu \phi \right] \right\} \delta g^{\mu\nu}
\end{aligned}$$

We identify the integrand as $\delta S_\phi / \delta g^{\mu\nu}$ and the energy-momentum tensor is then

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left[\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + V(\phi) \right]$$

The energy density $\rho = T_0^0$ and pressure $p \delta_j^i = T_j^i$ are

$$\rho = \frac{1}{2} \dot{\phi}^2 + V(\phi), \quad p = \frac{1}{2} \dot{\phi}^2 - V(\phi).$$

This is all nice, but let's move on to perturbations!!! Cosmological perturbations are cool since the large scale structure of our Universe can also be accounted for by inflation. Remember how we decomposed ϕ , likewise we can decompose the spacetime metric $g_{\mu\nu}$ as its background homogeneous and isotropic FLRW metric $g^{(0)}_{\mu\nu}(t)$, which is only time dependent, and the perturbation part $\delta g_{\mu\nu}$, which is spacetime dependent:

$$\begin{aligned}
\phi(x) &= \varphi(t) + \delta\phi(x) \\
g_{\mu\nu} &= g_{\mu\nu}^{(0)}(t) + \delta g_{\mu\nu}(x)
\end{aligned}$$

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Had a meeting today with Mindaugas, to determine the initial value for ϕ_i we basically can pick any N since it's relative and a free variable, like time. There are basically three stages. First we determine the initial values analytically, integrate until ϕ_{SBP} (which we ourselves pick) and then go a bit back and integrate with all the χ and θ fields. Also instead of using $b = \sqrt{\alpha}$ I should just use α . Another useful graph to make would be $\phi - \dot{\phi}$ graph.

Again, we look at slowroll. Using ϵ

$$\epsilon = \frac{1}{2} \left(\frac{V_{,\phi}}{V} \right)^2 = \frac{1}{2} \left(\frac{\kappa}{b} \right)^2 \exp \left(\frac{2\phi}{b} \right),$$

we know that at the end of inflation $\epsilon = 1$, hence we can determine the value of ϕ_{end}

$$\phi_{\text{end}} = \frac{b}{2} \ln \left(\frac{2b^2}{\kappa^2} \right)$$

then from the derived slowroll formula (7) assuming inflation ends at N_{end} we have

$$\begin{aligned} \phi_{\text{end}} &= -b \ln \left(e^{-\phi_i/b} + \frac{\kappa}{b^2} N_{\text{end}} \right) \\ \Rightarrow e^{-\phi_{\text{end}}/b} &= e^{-\phi_i/b} + \frac{\kappa}{b^2} N_{\text{end}} \\ \Rightarrow \phi_i &= -b \ln \left(e^{-\phi_{\text{end}}/b} + \frac{\kappa}{b^2} N_{\text{end}} \right) = -b \ln \left(\frac{\kappa}{\sqrt{2}b} - \frac{\kappa}{b^2} N_{\text{end}} \right). \end{aligned} \tag{10}$$

But notice that this puts a constraint on the possible values for N_{end} or b

$$\frac{\kappa}{b^2} N_{\text{end}} < \frac{\kappa}{\sqrt{2}b} \quad \Rightarrow \quad N_{\text{end}} < \frac{b}{\sqrt{2}} \quad \text{or} \quad b > \sqrt{2} N_{\text{end}}.$$

To find the initial value for ϕ' we first find the derivative and then similarly compute ϕ'_{end}

$$\phi'(N) = -\frac{\kappa}{b} \frac{1}{e^{-\phi_i/b} + \frac{\kappa}{b^2} N}$$

we can do a little trick by noticing that the denominator contains the same terms as the equation (7) hence using (10) we have

$$\phi'(N) = -\frac{\kappa}{b} \frac{1}{e^{\phi(N)/b}}$$

where I intentionally left the exponential in the denominator, for later convenience. Finally at $N = 0$ we have $\phi(N = 0) = \phi_i$, whose value we already determined. Plugging it in and cleaning up we're left with

$$\phi'(0) = \frac{1}{\frac{N_{\text{end}}}{b} - \frac{1}{\sqrt{2}}}$$

Now that we have all of the initial conditions we're able to integrate!!! For convenience here are all of the important formulas with $b = \sqrt{\alpha}$ inserted